

AI Cost Summary for Paper Extraction

Claude · GPT · Gemini — cost per paper, based on our real papers (Edexcel AS Physics, ~30 question-paper pages + ~19 mark-scheme pages) · 8 October 2026

How AI pricing works (in short)

Input = what the AI reads

- Our instructions + each **page picture** of the question paper and mark scheme.
- A page picture counts as about **1,100–1,550 tokens** (fixed by its size).
- One paper ≈ **130,000–150,000 input tokens**. Input is the cheaper part.

Output = what the AI writes

- Questions, LaTeX maths, marks, topics, answers and diagram positions.
- Top models also "think" before answering; this is billed as output.
- One paper ≈ **13,000 tokens** written + **10,000–30,000** thinking. Output costs more per token.

Diagrams cost nothing extra. The AI must see each diagram, but it sees it **inside the page picture** it is already reading, and a page is priced by its size, not by what is drawn on it. So a page with a diagram costs the same as a page without one (already included in the prices below). The AI only writes the diagram's position (~40 tokens, under \$0.02 per paper), and the picture is **cut out on our own server for free**. The cost grows with the **number of pages**, not the number of diagrams.

Price list (per 1 million tokens)

Tier	Model	Input	Output	Use
Top	Claude Opus 5.5	\$4.00	\$20.00	best quality
Top	GPT-5.5	\$5.00	\$30.00	best GPT quality
Top	Gemini 3.1 Pro	\$2.00	\$12.00	best Gemini quality
Mid	Claude Sonnet 5.5	\$2.00	\$10.00	strong, half of Opus
Mid	GPT-5.4	\$2.50	\$15.00	strong, half of GPT-5.5
Mid	Gemini 3.5 Flash	\$1.50	\$9.00	fast, good value
Low	Claude Haiku 4.5	\$1.00	\$5.00	cheap, weaker on hard pages
Low	GPT-5.4 mini	\$0.75	\$4.50	cheap
Low	GPT-5 mini	\$0.25	\$2.00	very cheap, weakest
Low	Gemini 3.5 Flash-Lite	\$0.30	\$2.50	very cheap, weakest

Cost per paper (typical paper: ~30 QP pages + ~19 MS pages)

Tier	Model	Per paper	100 papers	1,000 papers	10,000 papers
Top	Claude Opus 5.5	\$1.05–1.45	\$105–145	\$1,050–1,450	\$10.5k–14.5k
Top	GPT-5.5	\$1.35–1.95	\$135–195	\$1,350–1,950	\$13.5k–19.5k
Top	Gemini 3.1 Pro	\$0.55–0.75	\$55–75	\$550–750	\$5.5k–7.5k
Mid	Claude Sonnet 5.5	\$0.55–0.75	\$55–75	\$550–750	\$5.5k–7.5k
Mid	GPT-5.4	\$0.65–0.95	\$65–95	\$650–950	\$6.5k–9.5k
Mid	Gemini 3.5 Flash	\$0.40–0.60	\$40–60	\$400–600	\$4k–6k
Low	Claude Haiku 4.5	~\$0.20–0.25	\$20–25	\$200–250	\$2k–2.5k
Low	GPT-5.4 mini	\$0.20–0.30	\$20–30	\$200–300	\$2k–3k
Low	GPT-5 mini	\$0.08–0.12	\$8–12	\$80–120	\$800–1.2k
Low	Gemini 3.5 Flash-Lite	\$0.10–0.15	\$10–15	\$100–150	\$1k–1.5k

Ranges depend on how much the model "thinks". Batch mode (results within hours) is about 50% cheaper on all three providers.

Effect of paper size (top-tier models, mid thinking level)

Paper size	Claude Opus 5.5	GPT-5.5	Gemini 3.1 Pro
Small: 20 QP pages + 10 MS pages	~\$0.80	~\$1.05	~\$0.42
Typical: 30 QP pages + 19 MS pages (our papers)	~\$1.25	~\$1.65	~\$0.65
Large: 40 QP pages + 30 MS pages	~\$1.85	~\$2.45	~\$1.00

The number of diagrams does not change these figures; more pages cost more because every page is read. Claude prices are Anthropic's official prices; GPT and Gemini prices are from recent public pricing pages (October 2026) and should be confirmed on the official OpenAI and Google pages.

Rate limits (how fast we can send work to the AI)

Every AI provider limits how much an account can send in a given time. Going over the limit returns a "too many requests, please wait" error. Limits are not a cost, but they decide **how long an upload takes**.

Limit	Meaning
Requests per minute (RPM)	how many AI calls the account can make each minute
Tokens per minute (TPM)	how many tokens (input and/or output) per minute
Requests / tokens per day	daily cap, mainly on free keys

Claude (Anthropic)

Tier	Requests / min	Input tokens / min	Output tokens / min	How you reach it
Starting (Tier 1 / Start)	~50	~50,000	~10,000	first credit purchase (~\$5)
Tier 2	~1,000	higher	higher	more spend
Tier 3	~2,000	higher	higher	more spend
Tier 4	~4,000	much higher	much higher	higher spend

Since June 2026 Anthropic groups accounts as Start → Build → Scale (older guides use Tier 1–4). New accounts may also have a daily spend cap (around \$100/day reported). Batch mode has its own, larger limits.

ChatGPT / OpenAI (example: GPT-5.5)

Tier	Requests / min	Tokens / min	How you reach it
Tier 1	500	500,000	first payment
Tier 2	5,000	1,000,000	\$50 total spent
Tier 3	5,000	2,000,000	\$100 spent
Tier 4	10,000	4,000,000	\$250 spent
Tier 5	15,000	40,000,000	\$1,000 spent + 30 days

Tiers rise automatically with spend. A single GPT-5.5 request over 272k tokens is billed at a higher rate; ours are about 3–5k tokens per request.

Gemini (Google)

Tier	Requests / min	Tokens / min	Requests / day
Free	~5–15 (by model)	~250,000	~100–1,000
Tier 1 (paid)	~150–300	~1,000,000	~1,500
Tier 2	~500–1,500	~2,000,000	~10,000

Pro models (e.g. Gemini 3.1 Pro) are paid-only since April 2026. Google sets exact limits per project and shows them in the AI Studio dashboard.

What the limits mean for our papers

One paper ≈ 38 requests, ~150,000 input tokens and ~35,000 output tokens.

Account	Time per paper (limit-wise)	Papers per day (roughly)
Claude starting tier	~3–4 min (token limits)	~15 per hour, no daily request cap
Claude Tier 2+	~1–2 min	hundreds
OpenAI Tier 1	~1–2 min (limits are high enough)	hundreds
Gemini free (what we use now)	~3–8 min	only ~2–26 per day (daily request cap)
Gemini Tier 1 (paid)	~1–2 min	~39 per day (1,500 requests/day cap)
Gemini Tier 2	~1 min	~260 per day

Our app reads 3 pages at a time, so on any paid tier a paper takes a few minutes in practice. Limit figures are from Anthropic's rate-limit docs and recent public guides (October 2026); they change and depend on the account, so confirm them in each provider's console.

Our current setup

Item	Now	Effect
Upload AI	Gemini, free key	low limits and a daily cap; uploads can stop part-way through the day
App setting GEMINI_RPM	5 requests per minute	keeps us under the free limit
Requests per paper	~38 (28 question pages + 10 mark-scheme calls)	at 5 per minute: at least ~8 minutes per paper
When a limit is hit	app waits and retries, switches to backup models, skips and reports a failed page	the paper still completes

For real use

- Use a **paid key**: higher limits, a paper in a few minutes, no daily stop.
- Use **batch mode** for bulk uploads: about half price and separate, larger limits.
- For hundreds of papers, raise the account tier first or spread uploads over time.

Students are not affected

- The AI is used **only when an admin uploads a paper** (once per paper).
- Browsing, search, similar questions and PDF downloads run on our own server and database: **no AI tokens and no AI limits**.
- A visitor limit on our own server can be added later (e.g. in nginx) to stop abuse.